



CSE 201: Data Structure

Mathematical background

Few Maths

Mathematical background



Mathematics and engineering

For engineers, mathematics is a tool:

- Of course, that doesn't mean it always works...

$$\sqrt{\heartsuit} = ?$$

$$\cos \heartsuit = ?$$

$$\frac{d}{dx} \heartsuit = ?$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \heartsuit = ?$$

$$F\{\heartsuit\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{it\heartsuit} dt = ?$$

My normal approach
is useless here.



Justification

However, as engineers, you will not be paid to say:

Method A is *better* than Method B

or

Algorithm A is *faster* than Algorithm B

Such comparisons are said to be *qualitative*:

qualitative, *a.* Relating to, connected or concerned with, quality or qualities.
Now usually in implied or expressed opposition to
quantitative.



Justification

Qualitative statements cannot guide engineering design decisions:

- Algorithm A could be *better* than Algorithm B, but Algorithm A would require three person weeks to implement, test, and integrate while Algorithm B has already been implemented and has been used for the past year
- There are circumstances where it may be beneficial to use Algorithm A, but not based on the word *better*



Justification

Thus, we will look at a *quantitative* means of describing data structures and algorithms:

quantitative, a. Relating to, concerned with, quantity or its measurement;
ascertaining or expressing quantity.

This will be based on mathematics, and therefore we will look at a number of properties which will be used again and again throughout this course



Floor and ceiling functions

The *floor* function maps any real number x onto the greatest integer less than or equal to x :

$$\lfloor 3.2 \rfloor = \lfloor 3 \rfloor = 3$$

$$\lfloor -5.2 \rfloor = \lfloor -6 \rfloor = -6$$

- Consider it *rounding towards negative infinity*

The *ceiling* function maps x onto the least integer greater than or equal to x :

$$\lceil 3.2 \rceil = \lceil 4 \rceil = 4$$

$$\lceil -5.2 \rceil = \lceil -5 \rceil = -5$$

- Consider it *rounding towards positive infinity*



Logarithms

We will begin with a review of logarithms:

If $n = e^m$, we define

$$m = \ln(n)$$

It is always true that $e^{\ln(n)} = n$; however, $\ln(e^n) = n$ requires that n is real



Logarithms

Exponentials grow faster than any non-constant polynomial

$$\lim_{n \rightarrow \infty} \frac{e^n}{n^d} = \infty$$

for any $d > 0$

Thus, their inverses—logarithms—grow slower than any polynomial

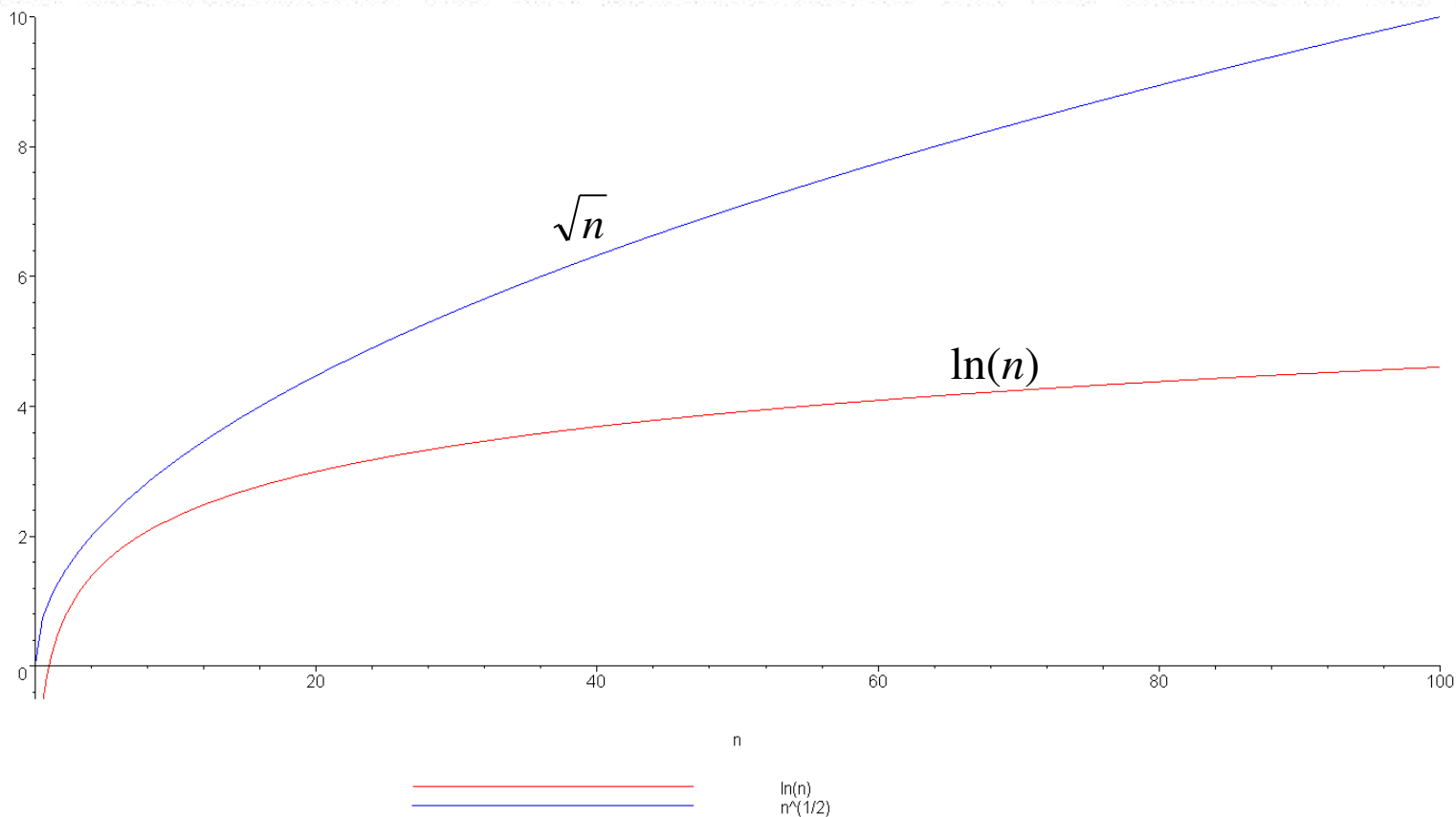
$$\lim_{n \rightarrow \infty} \frac{\ln(n)}{n^d} = 0$$



Logarithms

Example $f(n) = n^{1/2} = \sqrt{n}$

is strictly greater than $\ln(n)$

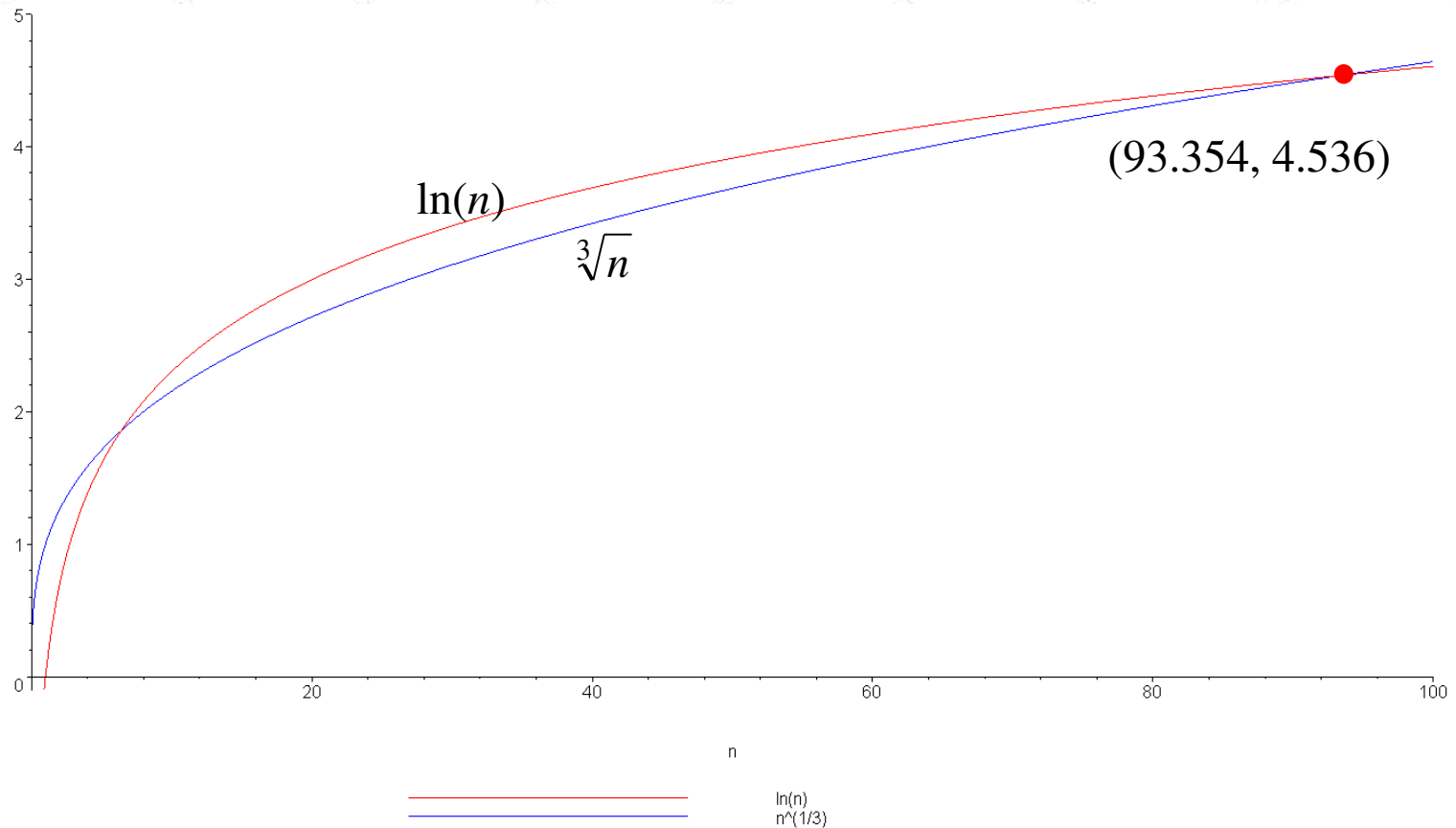




Logarithms

$$f(n) = n^{1/3} = \sqrt[3]{n}$$

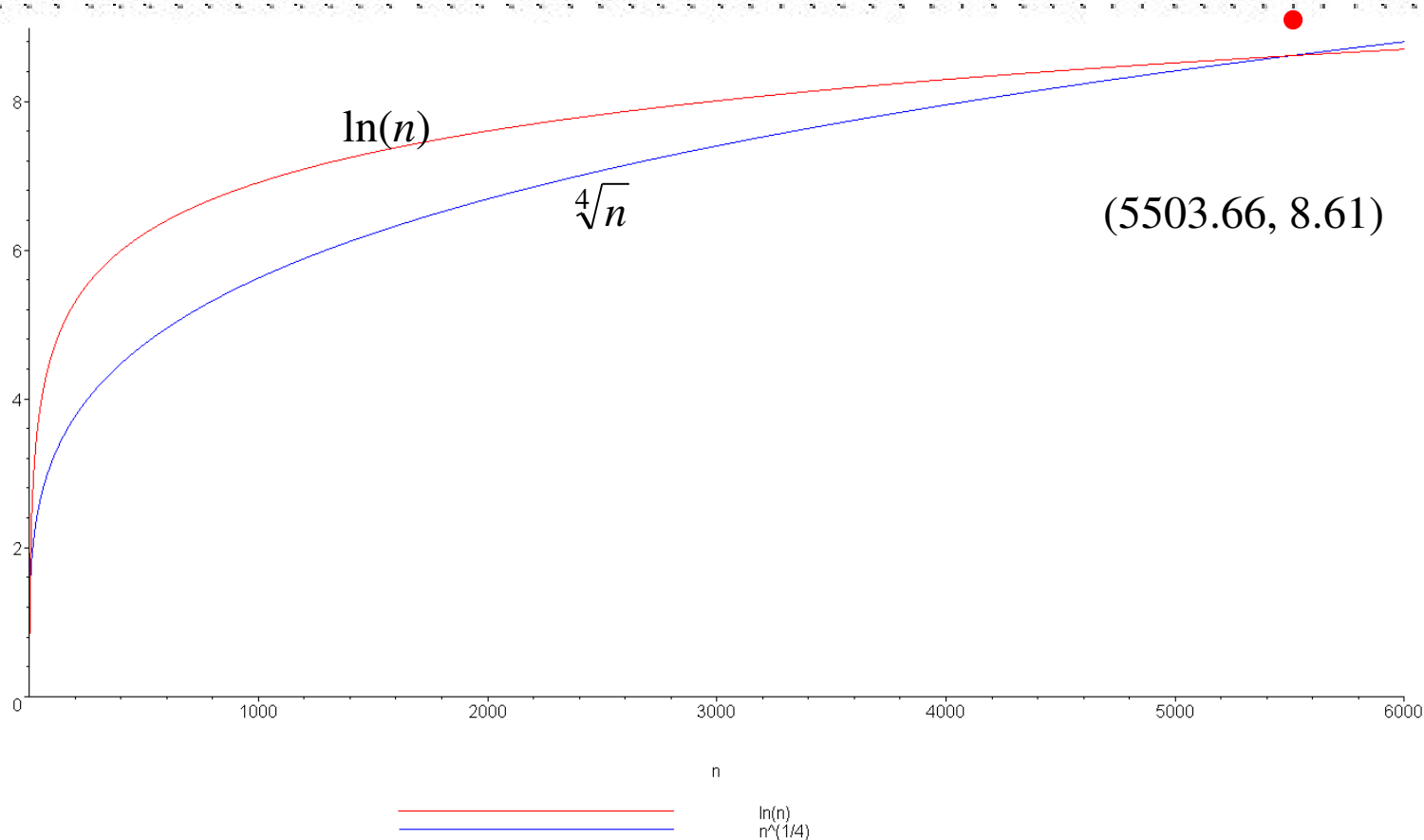
grows slower but only up to $n = 93$





Logarithms

You can view this with any polynomial





Logarithms

We have compared logarithms and polynomials

- How about $\log_2(n)$ versus $\ln(n)$ versus $\log_{10}(n)$

You have seen the formula

$$\log_b(n) = \frac{\ln(n)}{\ln(b)} = \left(\frac{1}{\ln(b)} \right) * \ln(n)$$

Constant

Because,

$$\log_b(n) = \frac{\log_a(n)}{\log_a(b)}$$

$$\log_{10}(n) = \frac{\ln(n)}{\ln(10)} = \left(\frac{1}{\ln(10)} \right) * \ln(n)$$

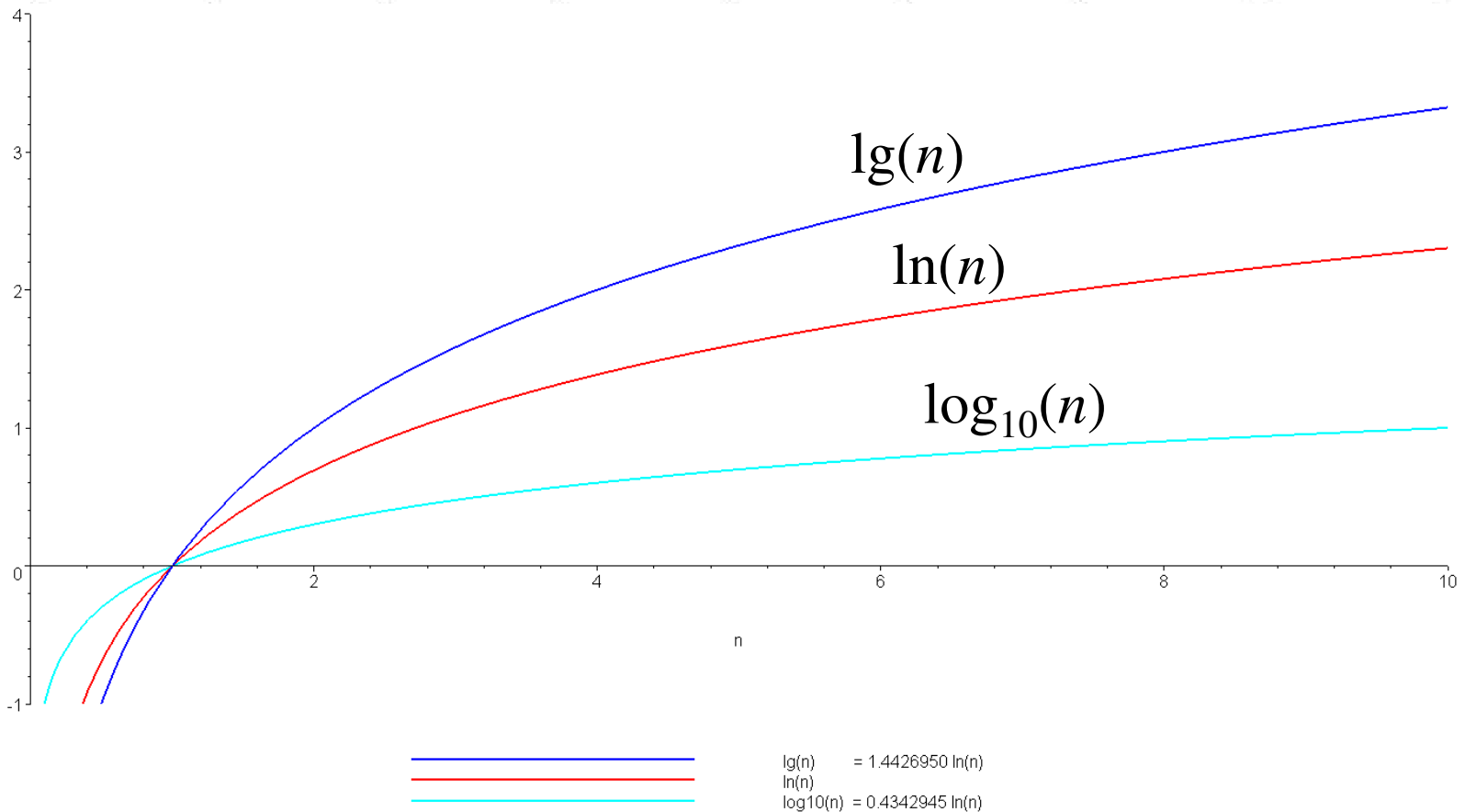
$$\log_2(n) = \frac{\ln(n)}{\ln(2)} = \left(\frac{1}{\ln(2)} \right) * \ln(n)$$

All logarithms are scalar multiples of each others



Logarithms

A plot of $\log_2(n) = \lg(n)$, $\ln(n)$, and $\log_{10}(n)$





Logarithms

You should also, as electrical or computer engineers be aware of the relationship:

$$\lg(2^{10}) = \lg(1024) = 10$$

$$\lg(2^{20}) = \lg(1\,048\,576) = 20$$

and consequently:

$$\lg(10^3) = \lg(1000) \approx 10 \quad \text{kilo}$$

$$\lg(10^6) = \lg(1\,000\,000) \approx 20 \quad \text{mega}$$

$$\lg(10^9) \approx 30 \quad \text{giga}$$

$$\lg(10^{12}) \approx 40 \quad \text{tera}$$



Arithmetic series

Next we will look various series

Each term in an arithmetic series is increased by a constant value (usually 1):

$$0 + 1 + 2 + 3 + \dots + n = \sum_{k=0}^n k = \frac{n(n+1)}{2}$$



Arithmetic series

Proof 1: write out the series twice and add each column

$$\begin{array}{ccccccccccccccc} 1 & + & 2 & + & 3 & + & \cdots & + & n-2 & + & n-1 & + & n \\ + & n & + & n-1 & + & n-2 & + & \cdots & + & 3 & + & 2 & + & 1 \\ \hline (n+1) & + & (n+1) & + & (n+1) & + & \cdots & + & (n+1) & + & (n+1) & + & (n+1) \end{array}$$
$$= n(n+1)$$

Since we added the series twice, we must divide the result by 2



Arithmetic series

Proof 2 (by induction):

The statement is true for $n = 0$:

$$\sum_{i=0}^0 k = 0 = \frac{0 \cdot 1}{2} = \frac{0(0+1)}{2}$$

Assume that the statement is true for an arbitrary n :

$$\sum_{k=0}^n k = \frac{n(n+1)}{2}$$



Arithmetic series

Using the assumption that

$$\sum_{i=0}^n i = \frac{n(n+1)}{2}$$

for n , we must show that

$$\sum_{k=0}^{n+1} k = \frac{(n+1)(n+2)}{2}$$



Arithmetic series

Then, for $n + 1$, we have:

$$\sum_{k=0}^{n+1} k = (n+1) + \sum_{i=0}^n k$$

By assumption, the second sum is known:

$$\begin{aligned} &= (n+1) + \frac{n(n+1)}{2} \\ &= \frac{(n+1)2 + (n+1)n}{2} \\ &= \frac{(n+1)(n+2)}{2} \end{aligned}$$



Arithmetic series

The statement is true for $n = 0$ and
the truth of the statement for n implies
the truth of the statement for $n + 1$.

Therefore, by the process of mathematical induction, the
statement is true for all values of $n \geq 0$.

Reference: Mr.

Oprendick



Other polynomial series

We could repeat this process, after all:

$$\sum_{k=0}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=0}^n k^3 = \frac{n^2(n+1)^2}{4}$$

however, it is easier to see the pattern:

$$\sum_{k=0}^n k = \frac{n(n+1)}{2} \approx \frac{n^2}{2}$$

$$\sum_{k=0}^n k^2 = \frac{n(n+1)(2n+1)}{6} \approx \frac{n^3}{3}$$

$$\sum_{k=0}^n k^3 = \frac{n^2(n+1)^2}{4} \approx \frac{n^4}{4}$$

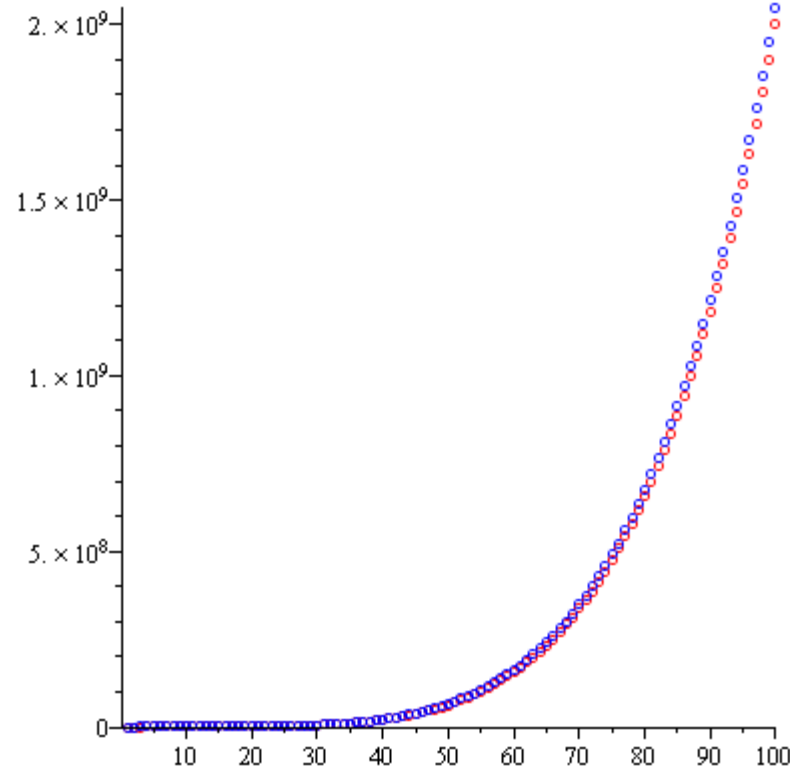
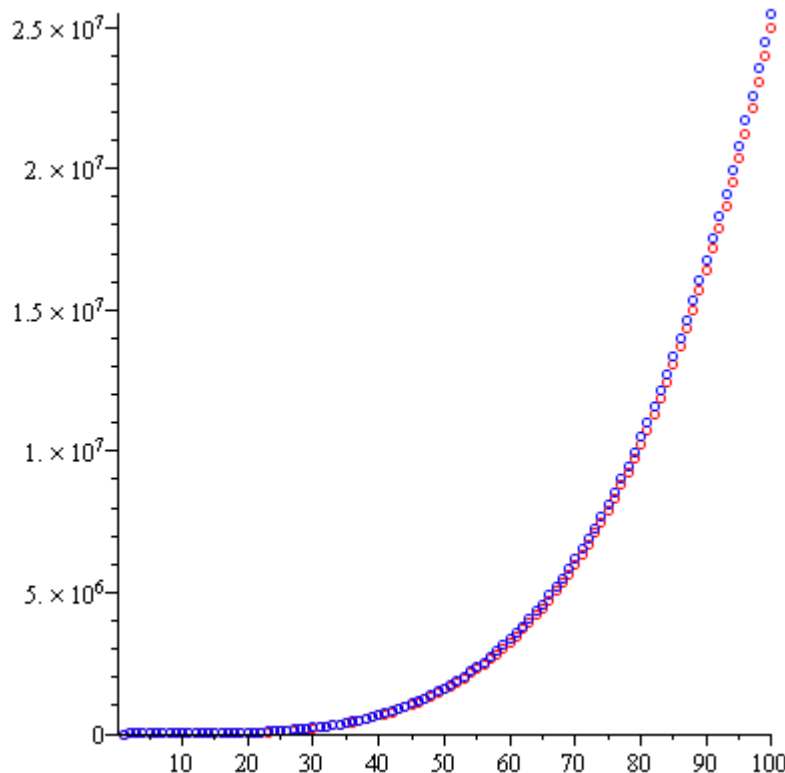


Other polynomial series

We can generalize this formula

$$\sum_{k=0}^n k^d \approx \frac{n^{d+1}}{d+1}$$

Demonstrating with $d = 3$ and $d = 4$:

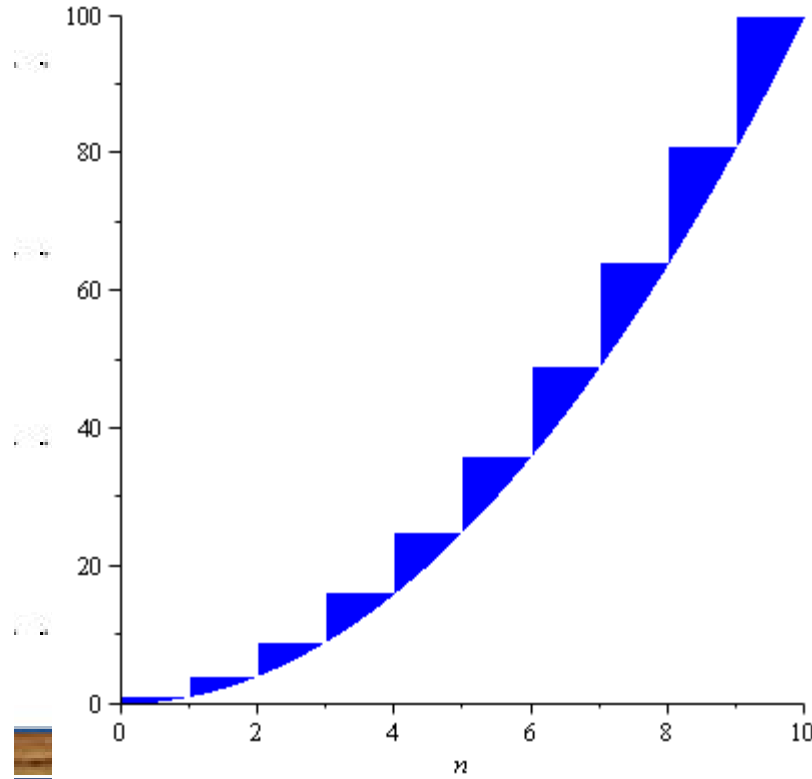
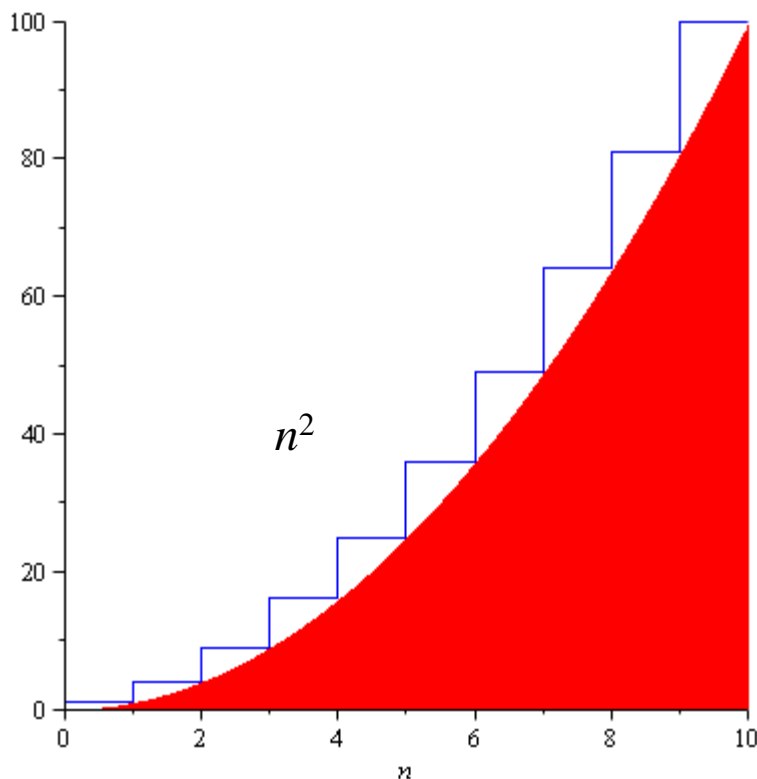




Other polynomial series

The justification for the approximation is that we are approximating the sum with an integral:

$$\sum_{k=0}^n k^d \approx \int_0^n x^d dx = \frac{x^{d+1}}{d+1} \Big|_{x=0}^n = \frac{n^{d+1}}{d+1} - 0$$





Other polynomial series

The ratio between the error and the actual value goes to zero:

- In the limit, as $n \rightarrow \infty$, the ratio between the sum and the approximation goes to 1

$$\lim_{n \rightarrow \infty} \frac{n^{d+1}}{\sum_{k=0}^n k^d} = 1$$

- *The relative error of the approximation goes to 0*



Geometric series

The next series we will look at is the geometric series with common ratio r :

$$\sum_{k=0}^n r^k = \frac{1 - r^{n+1}}{1 - r}$$

and if $|r| < 1$ then it is also true that

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1 - r}$$



Geometric series

Elegant proof: multiply by $\frac{1-r}{1-r}$

$$\sum_{k=0}^n r^k = \frac{(1-r) \sum_{k=0}^n r^k}{1-r}$$

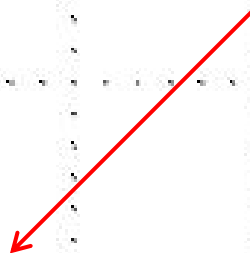
$$= \frac{\sum_{k=0}^n r^k - r \sum_{k=0}^n r^k}{1-r}$$

$$= \frac{(1 + r + r^2 + \dots + r^n) - (r + r^2 + \dots + r^n + r^{n+1})}{1-r}$$

$$= \frac{1 - r^{n+1}}{1-r}$$

Telescoping series:

all but the first and last terms cancel



Ref: Bret D. Whissel, *A Derivation of*



Geometric series

Note that we can use a change-of-index with summations like we do with integration:

$$\sum_{i=1}^n r^i = \sum_{i=1}^n r r^{i-1} = r \sum_{i=1}^n r^{i-1}$$

Letting $j = i - 1$:

$$= r \sum_{j=0}^{n-1} r^j = r \frac{1 - r^n}{1 - r}$$



Recurrence relations

Finally, we will review recurrence relations:

- Sequences may be defined explicitly: $x_n = 1/n$

$$1, 1/2, 1/3, 1/4, \dots$$

Here n is x_n independent.

- Consider a fibonacci series where $F_1=1$, $F_2=2$ and $F=F+F$

Then the series is 1, 2, 3, 5, 8, 13, 21, ...

- A recurrence relationship is a means of defining a sequence based on previous values in the sequence
- Such definitions of sequences are said to be *recursive*



Recurrence relations

Define an initial value: e.g., $x_1 = 1$

Defining x_n in terms of previous values:

- For example,

$$x_n = x_{n-1} + 2$$

$$x_n = 2x_{n-1} + n$$

$$x_n = x_{n-1} + x_{n-2}$$



Recurrence relations

Given the two recurrence relations

$$x_n = x_{n-1} + 2$$

$$x_n = 2x_{n-1} + n$$

and the initial condition $x_1 = 1$ we would like to find explicit formulae for the sequences

In this case, we have

$$x_n = 2n - 1$$

$$x_n = 2^{n+1} - n - 2$$

respectively



Recurrence relations

We will use a functional form of recurrence relations:

Calculus

$$x_1 = 1$$

$$x_n = x_{n-1} + 2$$

$$x_n = 2x_{n-1} + n$$

CSE

$$f(1) = 1$$

$$f(n) = f(n-1) + 2$$

$$f(n) = 2 f(n-1) + n$$



Recurrence relations

The previous examples using the functional notation are:

$$f(n) = f(n-1) + 2$$

$$g(n) = 2g(n-1) + n$$

With the initial conditions $f(1) = g(1) = 1$, the solutions are:

$$f(n) = 2n - 1$$

$$g(n) = 2^{n+1} - n - 2$$



Recurrence relations

In some cases, given the recurrence relation, we can find the explicit formula:

- Consider the Fibonacci sequence:

$$f(n) = f(n-1) + f(n-2)$$

$$f(0) = f(1) = 1$$

that has the solution

$$f(n) = \frac{2 + \phi}{5} \phi^n + \frac{3 - \phi}{5} \phi^{-n}$$

where ϕ is the golden ratio:

$$\phi = \frac{\sqrt{5} + 1}{2} \approx 1.6180$$



Weighted averages

Given n objects $x_1, x_2, x_3, \dots, x_n$, the *average* is

$$\frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

Given a sequence of coefficients $c_1, c_2, c_3, \dots, c_n$ where

$$c_1 + c_2 + c_3 + \dots + c_n = 1$$

then we refer to

$$c_1 x_1 + c_2 x_2 + c_3 x_3 + \dots + c_n x_n$$

as a *weighted average*

For an average $c_1 = c_2 = c_3 = \dots = c_n = \frac{1}{n}$



Weighted averages

Examples:

- Simpson's method approximates an integral by sampling the function at three points: $f(a)$, $f(b)$, $f(c)$
- The average value of the function is approximated by

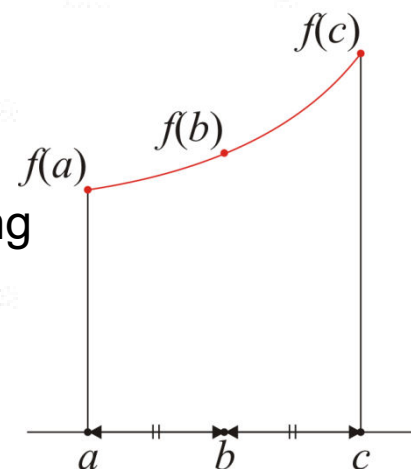
$$\frac{1}{6} f(a) + \frac{2}{3} f(b) + \frac{1}{6} f(c)$$

- It can be shown that that

$$\left(\frac{1}{6} f(a) + \frac{2}{3} f(b) + \frac{1}{6} f(c) \right) (c - a)$$

is a significant better approximation than

$$\frac{f(a) + f(b) + f(c)}{3} (c - a)$$





Weighted averages

Examples:

$$\int_0^2 \cos(x) dx = \sin(2) \approx 0.9093$$

– Using the weighted average:

$$\left(\frac{1}{6} \cos(0) + \frac{2}{3} \cos(1) + \frac{1}{6} \cos(2) \right) \times 2 \approx 0.9150$$

$$\frac{1}{6} + \frac{2}{3} + \frac{1}{6} = 1$$

– Using a simple average:

$$\frac{\cos(0) + \cos(1) + \cos(2)}{3} \times 2 \approx 0.7494$$

$$\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$



Combinations

Given n distinct items, in how many ways can you choose k of these?

- I.e., “In how many ways can you combine k items from n ?”
- For example, given the set $\{1, 2, 3, 4, 5\}$, I can choose three of these in any of the following ways:

$\{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}, \{1, 3, 4\}, \{1, 3, 5\},$
 $\{1, 4, 5\}, \{2, 3, 4\}, \{2, 3, 5\}, \{2, 4, 5\}, \{3, 4, 5\}$

The number of ways such items can be chosen is written

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

where $\binom{n}{k}$

is read as “ n choose k ”s

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

There is a recursive definition:



Combinations

The most common question we will ask in this vein:

- Given n items, in how many ways can we choose two of them?
- In this case, the formula simplifies to:

$$\binom{n}{2} = \frac{n!}{2!(n-2)!} = \frac{n(n-1)}{2}$$

For example, given $\{0, 1, 2, 3, 4, 5, 6\}$, we have the following 21 pairs:

$\{0, 1\}, \{0, 2\}, \{0, 3\}, \{0, 4\}, \{0, 5\}, \{0, 6\},$
 $\{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{1, 6\},$
 $\{2, 3\}, \{2, 4\}, \{2, 5\}, \{2, 6\},$
 $\{3, 4\}, \{3, 5\}, \{3, 6\},$
 $\{4, 5\}, \{4, 6\},$
 $\{5, 6\}$



Combinations

You have also seen this in expanding polynomials:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

For example,

$$(x + y)^4 = \sum_{k=0}^4 \binom{4}{k} x^k y^{4-k}$$

$$= \binom{4}{0} y^4 + \binom{4}{1} x y^3 + \binom{4}{2} x^2 y^2 + \binom{4}{3} x^3 y + \binom{4}{4} x^4$$

$$= y^4 + 4xy^3 + 6x^2y^2 + 4x^3y + x^4$$



Combinations

These are also the coefficients of Pascal's triangle:

$$\begin{array}{ccccccccccc} & & & & \binom{0}{0} & & & & & & \\ & & & \binom{1}{0} & & \binom{1}{1} & & & & & \\ & & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & & & & \\ & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & & & \\ \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4} & & \end{array}$$
$$\begin{array}{ccccccccccc} & & & & & & & & & & \\ & & & & & & & & & 1 & \\ & & & & & & & & 1 & & 1 \\ & & & & & & 1 & & 2 & & 1 \\ & & & & 1 & & 3 & & 3 & & 1 \\ & & 1 & & 4 & & 6 & & 4 & & 1 \end{array}$$



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Mathematical background

Few Maths

Absolute Value or Modulus of a

$|a|$ = non-negative value of a

Example:

$$|5| = 5$$

$$|0| = 0$$

$$|-5| = 5$$

$$|-5.5| = 5.5$$



Modulo

the result of the modulo operation is the remainder of the [Euclidean division](#).

In nearly all computing systems, the [quotient](#) q and the remainder r of a divided by n satisfy

q is an integer number.

$$a = nq + r$$

$$|r| < |n|$$

This is then written as

$$r = (a \bmod n)$$

- Identity:

- $(a \bmod n) \bmod n = a \bmod n$.
- $n^x \bmod n = 0$ for all positive integer values of x .
- If p is a [prime number](#) which is not a [divisor](#) of b , then $ab^{p-1} \bmod p = a \bmod p$, due to [Fermat's little theorem](#).

Inverse:

- $[(-a \bmod n) + (a \bmod n)] \bmod n = 0$.
- $b^{-1} \bmod n$ denotes the [modular multiplicative inverse](#), which is defined if and only if b and n are [relatively prime](#), which is the case when the left hand side is defined: $[(b^{-1} \bmod n)(b \bmod n)] \bmod n = 1$.



Modulo

- Identity:

- $(a \bmod n) \bmod n = a \bmod n$.
- $n^x \bmod n = 0$ for all positive integer values of x .
- If p is a prime number which is not a divisor of b , then $ab^{p-1} \bmod p = a \bmod p$, due to Fermat's little theorem.

Inverse:

- $[(-a \bmod n) + (a \bmod n)] \bmod n = 0$.
- $b^{-1} \bmod n$ denotes the modular multiplicative inverse, which is defined if and only if b and n are relatively prime, which is the case when the left hand side is defined: $[(b^{-1} \bmod n)(b \bmod n)] \bmod n = 1$.

- Distributive:

- $(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$.
- $ab \bmod n = [(a \bmod n)(b \bmod n)] \bmod n$.

- Division (definition): $\frac{a}{b} \bmod n = [(a \bmod n)(b^{-1} \bmod n)] \bmod n$, when the right hand side is defined

- Inverse multiplication: $[(ab \bmod n)(b^{-1} \bmod n)] \bmod n = a \bmod n$.

$$(-a \bmod n) = n - (a \bmod n)$$

Example:

$$16 \bmod 3 = 1, \quad -16 \bmod 3 = 2$$



End of the Chapter